

Content Coverage of PNAS in 1982-2001

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This paper reports results of determining changes in content coverage of the Proceedings of the National Academy of Science (PNAS) for the years 1982-2001. Dominant keywords were extracted to visualize major research topics and their evolution in the 20-year time frame. Two time slices namely 1995 and 2001 are examined in detail. Latent Semantic Analysis [6] was applied to determine the space of semantically similar documents for both years. The results of a co-word analysis of terms used in titles and keywords of articles were visualized using graph layout algorithms available in the graphical visualization software *Pajek* [1].

Data Set

The utilized data set comprises the complete set of papers in the Proceedings of National Academy of Science

for the years 1982-2001. For each year the top 20 words taken from titles, publisher assigned keywords and controlled vocabulary Mesh terms were extracted.

Subsequently, the number of resulting terms was examined for major trends and eight keywords were selected for closer examination. Figure 1 plots the frequency of these words for years in which they were among the top 20 most frequently used words together with key events. In addition, Kleinberg's burst detection algorithm [2] was applied to visualize bursts – major increases in the usage frequency – of those eight words.

Correlating the graph with historical biotechnology events reported in [3] results in the interpretation below. Human studies are steadily increasing and burst occurred by the start of the Human Genome Project in 1988. The frequency of Transcription is coupled with the PCR technology conceived in 1983 and fully developed by 1986. The application of PCR in combination with reverse

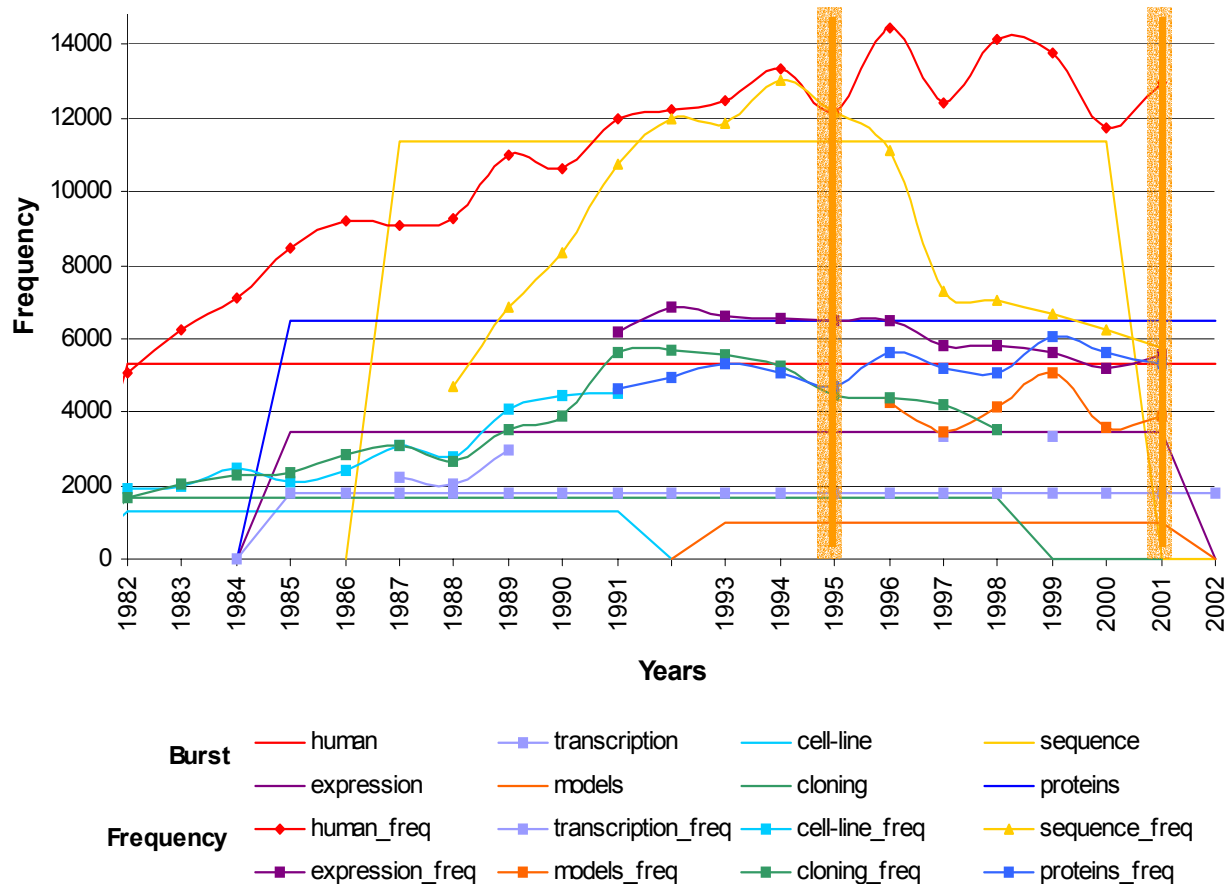


Figure 1: Bursts and Frequencies of the Top Eight Keywords

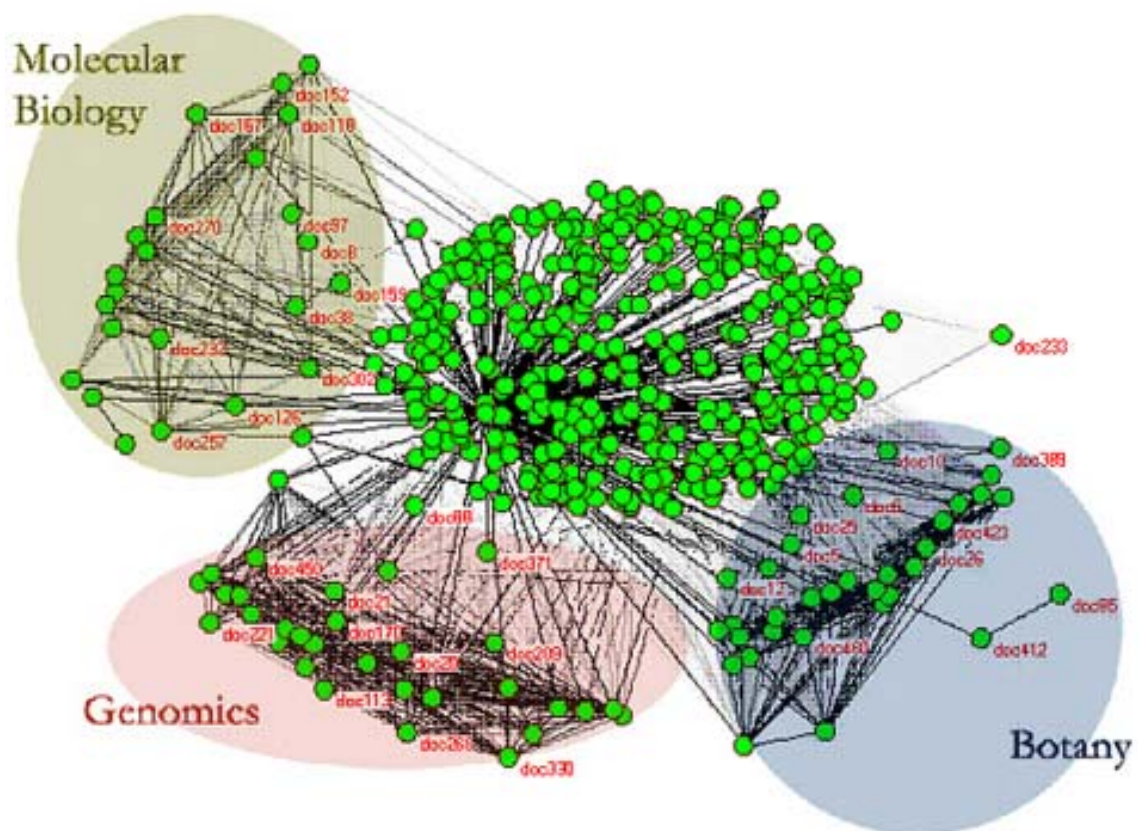


Figure 2: Document Space for 1995

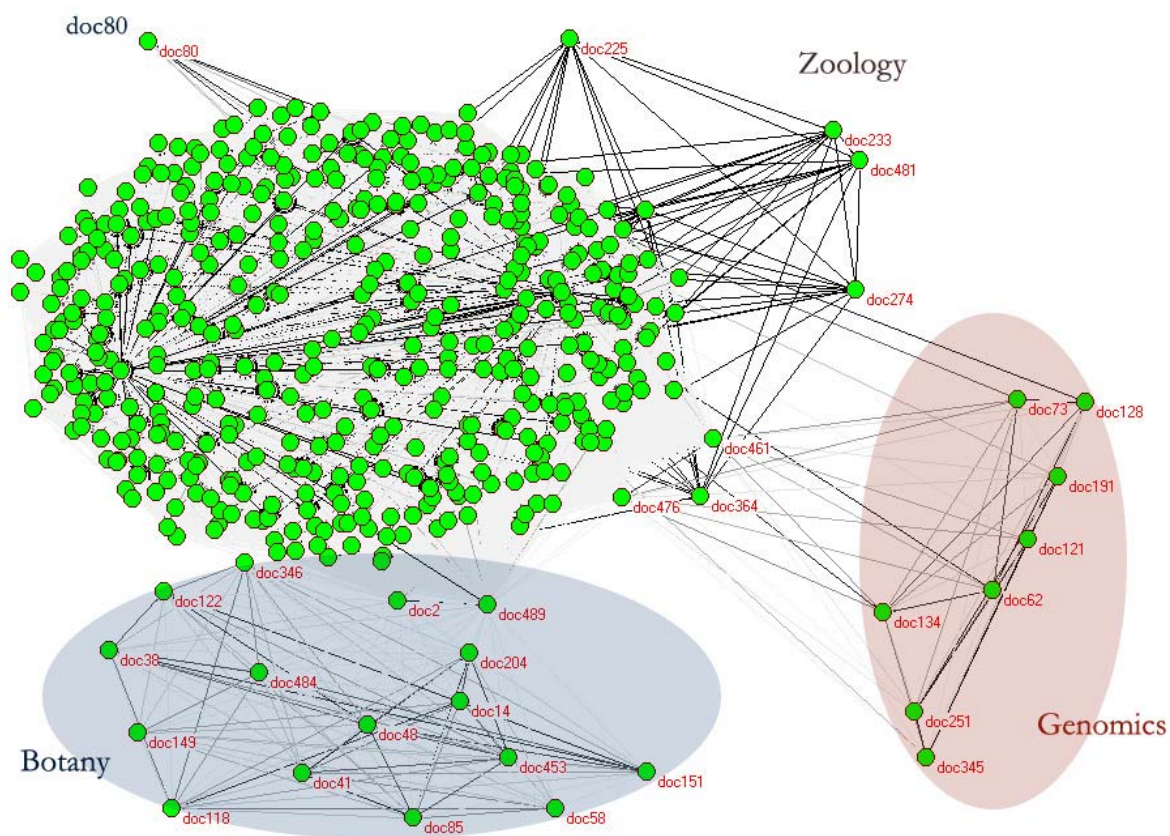


Figure 3: Document Space for 2001

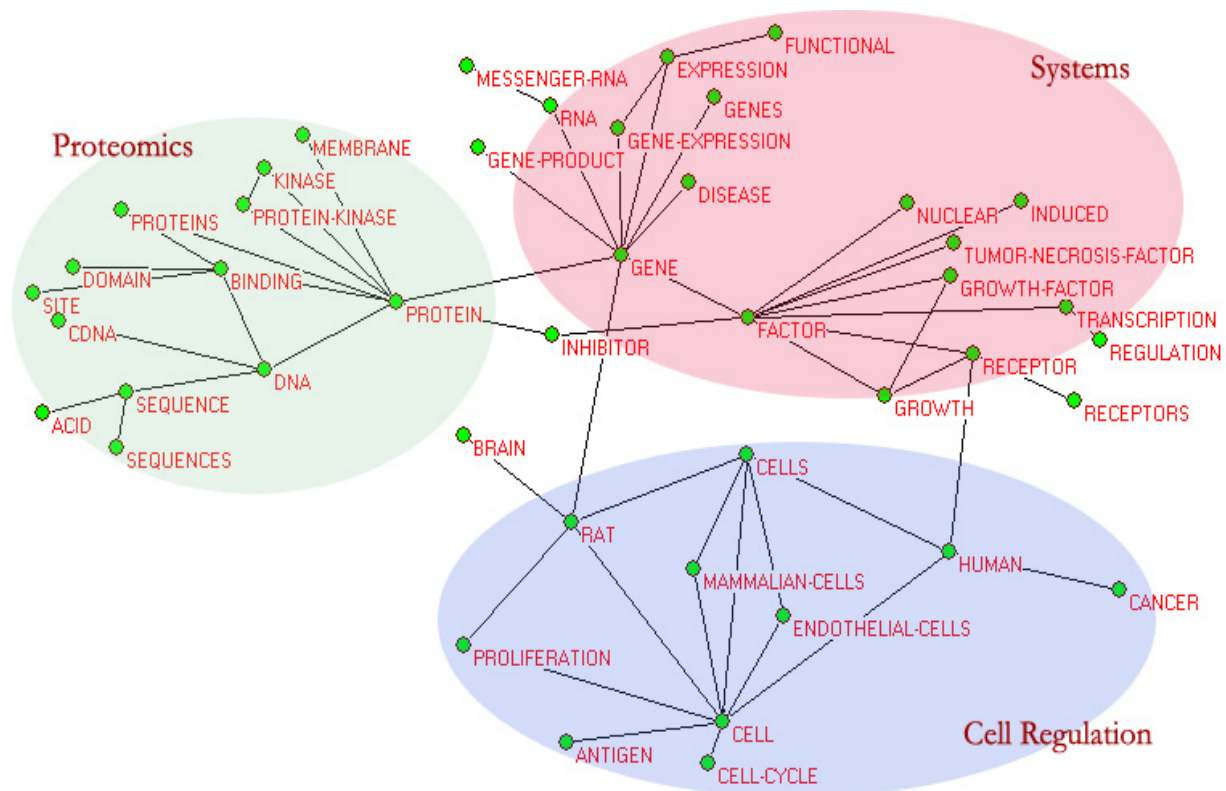


Figure 4: PNAS (key)word space 1995

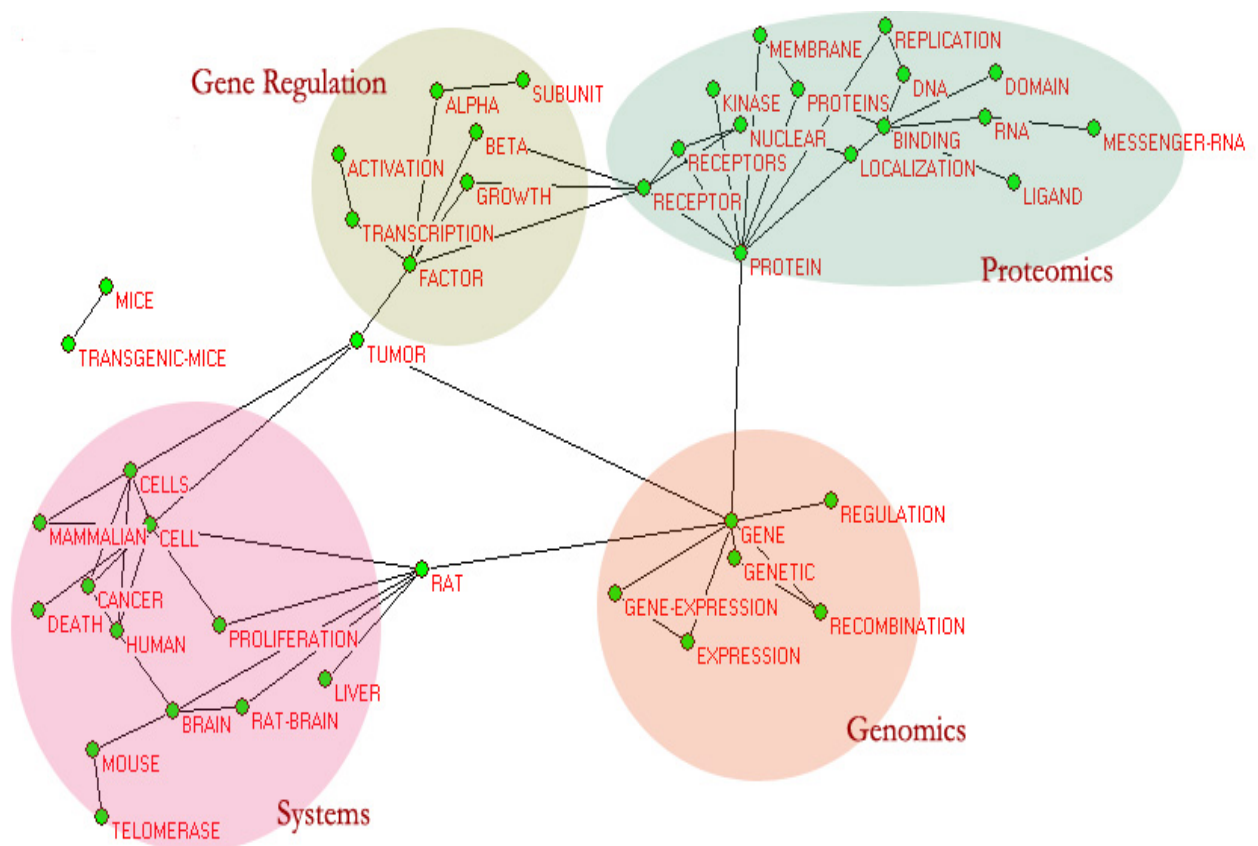


Figure 5: PNAS (key)word space 2001

transcription was increased in related papers starting in 1986 until research concentrated on gene expression experiments. Cell Line, a term frequently used 1982-1991 is primarily associated with the tissue cultures which quickly become routine. Molecular Sequence Data research also received a boost with sanction of funds for Human Genome Project. However it starts declining after 1994 as it becomes routine process in genetic studies. In 1991, work on the Expression rate of DNA to protein conversion (rather than just the transcription process) takes off. As more data on molecular sequence, expression rate, etc. become available research towards developing molecular models and protein modeling becomes relevant. Cloning is among the top 10 keywords for 17 years and research results in the success of cloning of a sheep in 1997. It also combines with wide spread use of molecular cloning approaches adopted in the previous years. Co-occurrence of *Proteins* and *Expression* is relevant as the research is tightly coupled.

Interested to get a more detailed view of research fields and term interrelationships, two years – 1995 and 2001 – were examined in more detail. For each document the author and publisher assigned keywords and controlled vocabulary Mesh terms from Medline were determined. Statistics on the two datasets are given in Table 1. Obviously papers published in 2001 had little time to receive a significant number of citations.

Table 1: Statistics of the PNAS data sets

Features	1995	2001
Number of documents	2505	2708
Number of unique keywords	14408	15269
Average number of keywords	5.8	5.6
Maximum number of times cited	1447	155

In order to make the data set more manageable, the 500 most cited articles were selected for the subsequent analysis of content coverage. This results in a data set for 1995 in which each document was at least 90 times cited and in the 2001 data set each document received at least 10 citations.

Data Analysis

Latent semantic analysis (LSA), also called latent semantic indexing was applied to determine sets of semantically similar documents [4]. LSA extends the vector space model by modeling term-document relationships using a reduced approximation for the column and row space computed by the singular value decomposition of the term by document matrix [5]. By considering the context of the words, LSA overcomes two fundamental problems faced by traditional lexical matching schemes: synonymy (similar meaning words – impacts recall) and polysemy (words with multiple meaning – impacts precision).

Using code for data parsing and similarity matrix computation available in the information-visualization

repository¹ at Indiana University the list of unique terms and the term-by-document frequency matrix were determined for the 1995 and 2001 data set. Subsequently, the LSA SVDPACKC provided by M. Berry [6] was applied to determine the most important latent dimensions. The document-by-document similarity matrices were computed based on the 118 most important dimensions for 1995 and 136 dimensions for 2001.

The word co-occurrence space was created based on the original list of keywords as well as words occurring in the titles of publications.² Stop words were eliminated as well as words that occurred less than ten times. The word-by-word similarity matrix was generated based on the co-occurrence of (key)word in the 500 most cited documents for each year. For example, if PROTEIN and GENE are used as keywords in one document then their frequency value is increased by one and hence the similarity increases. The values of the frequency matrices generated for each year were divided by the highest value 271 (1995) and 222 (2001) to obtain the word-by-word similarity matrices.

Data Visualization

A) Semantic Document Space

The Kamada Kawai algorithm [7] implemented in Pajek [1] was used to layout the documents in a 2-dimensional space. The results are displayed in Figure 2 and 3 and are discussed subsequently.

The visualization of the 500 most cited documents published in 1995 shows four major clusters. Three clusters have been labeled *Genomics*, *Botany* and *Molecular Biology* after a careful examination of documents in those clusters. Doc233 (original ID is A1995RB80400002) interconnects different areas with the keywords:

+BREAST-CANCER|+CELL-PROLIFERATION|+COLON
CANCER|+DIETARY RESTRICTION|+EPIDEMIOLOGIC
EVIDENCE|+HEMATOLOGICAL FINDINGS|+HEPATITIS-
CVIRUS|+HEPATOCELLULAR-CARCINOMA|+LOW-
INCOMEHOUSEHOLDS|+NON-HODGKINS-LYMPHOMA

The visualization of the 500 most cited papers published in 2001 shows three major clusters. Two of these clusters contain documents describing research in *Genomics* and *Botany*. The more densely packed middle cluster indicates a higher interconnectedness of the papers under study.

Doc 274, 481, 364, 225 (upper right corner) discuss research in *Zoology*. Doc 80 is of special interest as its keywords ‘biosynthesis’ and ‘insect herbivores’ bridge between different domains.

B) Co-(Key) Word Space

Figure 4 and 5 show the co-(key)word spaces visualized in Pajek using the Fruchterman-Reingold 2D-algorithm [8]. The visualization shows distinct cluster formations of the hot-topics covered in the years under consideration. In

¹ <http://iv.slis.indiana.edu/>

² In biological research it is common practice that words occurring in the title cannot be used as keywords. Hence titles that accurately describe the content of a paper imply keywords that may not perfectly fit.

consultation with a genomics research expert major clusters have been identified.

As for 1995, displayed in Figure 4, three major clusters exist: *Proteomics* (green), *Cell Regulation* (blue) and *Systems* (red).

The 2001 keyword coverage in Figure 5 shows similar topic coverage but with increasing specialization of research. While the *Systems* cluster in 1995 covered the genomics research, in 2001 *Genomics* (rose) is now a separate field and more specific clusters like *Gene Regulation* (brown) exist.

A visual comparison of the plots leads to the conclusion that there are similar patterns in the usage of keywords in the two years under study. Interestingly, documents appear to form larger clusters in 2001 (large middle cluster) whereas the (key)word space shows an increasing specialization.

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